Capacitated Vehicle Routing Quantum Clustering Optimization Using Analog Neutral Atoms with Compact Slack-Free Encoding

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1. Overview

The Capacitated Vehicle Routing Problem (CVRP) is a combinatorial optimization problem in which customers must be assigned to a fleet of capacity-constrained vehicles while minimizing total transportation cost [1]. Classical formulations become computationally demanding as problem size increases, motivating the investigation of quantum optimization methods. Quadratic Unconstrained Binary Optimization (QUBO) provides a framework for representing combinatorial objectives and constraints as quadratic functions of binary variables, which can subsequently be mapped to Ising Hamiltonians [2, 3].

A central challenge in implementing constrained optimization on neutral-atom quantum processors is the limited number of available qubits and the restricted form of physically realizable interactions. Conventional QUBO formulations often introduce auxiliary slack variables to encode inequality constraints, increasing the number of required qubits and interactions. Recent research has explored slack-free penalty formulations to reduce this overhead [4, 5]. Neutral-atom architectures introduce further restrictions through their distance-dependent Rydberg interactions and limited detuning control [6].

We investigate a reduced-variable QUBO formulation for capacity-constrained clustering, corresponding to Track A of the Pasqal Quantum Hackathon. For Tier S, consisting of four customers and two vehicles, we encode each customer assignment using a single binary variable, requiring only four atoms. Capacity constraints are approximated through a quadratic load-balancing penalty rather than explicit slack variables. The resulting QUBO is embedded into a neutral-atom register and evaluated through analog quantum emulation using prescribed Rabi-frequency and detuning waveforms. We assess solution feasibility, optimality gaps, and embedding fidelity against exact classical references, including the CVRP cost obtained by routing feasible clusters exactly.

2. Tier-S Mathematical Model

2.1. Problem Definition

We consider the Tier S capacitated vehicle routing problem, consisting of four customers and two identical vehicles, each with capacity seven. Customer locations are represented by Euclidean coordinates, with pairwise distances defined as

$$N = 4, \quad K = 2, \quad Q = 7, \quad D_{ij} = |\mathbf{r}_i - \mathbf{r}_j|_2.$$

Each customer has demand $d_i \in \{1, 2, 3, 4\}$ and is represented by a single binary assignment variable:

$$z_i = \begin{cases} 1 & \text{if assigned to vehicle 1,} \\ 0 & \text{if assigned to vehicle 2.} \end{cases}$$

This encoding guarantees that each customer is assigned to exactly one vehicle without additional assignment constraints.

We minimize the total pairwise distance between customers assigned to the same vehicle [6]:

$$\boxed{C(z) = \sum_{i<j} D_{ij}(1 - z_i - z_j + 2z_i z_j)} \tag{1}$$

The corresponding constrained optimization problem is

$$\begin{aligned} \min_{z \in 0,1^N} \quad & C(z) \text{ subject to} \\ & \sum_i d_i z_i \le Q, \qquad \sum_i d_i (1 - z_i) \le Q. \end{aligned} \tag{2}$$

2.2. Slack-Free Capacity Penalization

Conventional QUBO formulations encode capacity inequalities using binary slack variables, increasing the number of required qubits and interactions [2, 3, 8]. Motivated by research on unbalanced penalization [4], we instead approximate the capacity constraints using a quadratic load-balancing surrogate.

Define the total demand and capacity margins:

$$\begin{aligned} T &= \sum_i d_i, \; h_1(z) \\ &= Q - \sum_i d_i z_i, \; h_2(z) \\ &= Q - \sum_i d_i (1 - z_i). \end{aligned} \tag{3}$$

Consider the penalty function

$$P(z) = -\alpha(h_1 + h_2) + \lambda(h_1^2 + h_2^2), \qquad \lambda > 0. \tag{4}$$

Our penalty formulation is motivated by the unbalanced penalization approach of Montañez-Barrera et al. [4], which approximates inequality constraints through linear and quadratic functions of their capacity margins. However, in our two-vehicle formulation, the sum of the linear terms is constant, leaving a quadratic penalty that favors balanced vehicle loads rather than directly penalizing constraint violations.

Since the sum of the capacity margins is constant,

$$h_1 + h_2 = 2Q - T,$$

the linear penalty contributes only a constant offset. Expanding the quadratic terms and removing constants yields

$$\boxed{P(z) = 2\lambda\left(\sum_i d_i z_i - \frac{T}{2}\right)^2} \tag{5}$$

The complete unconstrained objective is therefore

$$\boxed{\begin{aligned} F(z) = \sum_{i<j} D_{ij}(1 - z_i - z_j + 2z_i z_j) \\ +2\lambda\left(\sum_i d_i z_i - \frac{T}{2}\right)^2. \end{aligned}} \tag{6}$$

The penalty weight controls the tradeoff between geographic clustering and demand balancing.

Unlike an exact inequality penalty, this surrogate does not guarantee capacity feasibility. Consequently, measured assignments are independently checked against the original constraints in Equation (2), and only feasible assignments are included in the reported optimality gaps.

We select $\lambda = 0.1$ as a relatively small penalty weight to encourage load balancing without allowing the penalty to dominate the geographic clustering objective. This choice is heuristic rather than a theoretically guaranteed constraint-enforcement threshold. Its effectiveness is assessed empirically through capacity feasibility and optimality gaps across generated instances.

2.3. QUBO Representation

Expanding Equation (6) into standard QUBO form [2] gives

$$\boxed{F(z) = c + z^{\mathsf{T}} M z} \tag{7}$$

where the symmetric matrix M has diagonal coefficients

$$M_{ii} = -\sum_{j \neq i} D_{ij} + 2\lambda(d_i^2 - T d_i) \tag{8}$$

and off-diagonal coefficients

$$M_{ij} = D_{ij} + 2\lambda d_i d_j, \qquad i \neq j. \tag{9}$$

The constant offset is

$$c = \sum_{i<j} D_{ij} + \frac{\lambda T^2}{2}. \tag{10}$$

This offset can be omitted during optimization without changing the minimizing assignments.

The off-diagonal coefficients are strictly positive for positive demands and penalty weight, consistent with repulsive Rydberg interactions. However, the generally unequal diagonal coefficients introduce a challenge for neutral-atom implementations restricted to global detuning.

2.4. Variable Efficiency

A conventional one-hot assignment encoding requires NK binary variables. Binary-encoded capacity slack introduces an additional $K\lceil \log_2(Q+1) \rceil$ variables.

Encoding	Tier S variables
One-hot assignment	8
One-hot assignment with binary slack	14
Proposed slack-free formulation	**4**

Our formulation reduces the binary-variable requirement from 14 to 4 relative to the conventional one-hot, slack-variable formulation.

$$\boxed{N_{\text{bits}} = N = 4} \tag{11}$$

This reduction enables the clustering QUBO to be represented using only four neutral atoms, at the cost of approximating rather than exactly enforcing the capacity constraints.

3 Quantum Implementation and Emulation

3.1. Hamiltonian Mapping and Embedding

We map each binary variable to a neutral atom, where $|0\rangle = |g\rangle$ and $|1\rangle = |r\rangle$. Following the Rydberg-based QUBO implementation described in Pulser [8], the system evolves under the Hamiltonian

$$\frac{\Omega(t)}{2}\sum_i X_i - \delta(t)\sum_i \hat{n}_i + \sum_{i<j}\frac{\widetilde{C}_6}{R_{ij}^6}\hat{n}_i\hat{n}_j, \tag{12}$$

where $\Omega(t)$ is the Rabi frequency, $\delta(t)$ is the global detuning, $\hat{n}_i = |r\rangle_i\langle r|$ is the excitation-number operator, and R_{ij} is the interatomic distance.

To approximate the QUBO interactions, we introduce a positive energy-scaling factor γ and match the physical interaction strengths to the quadratic coefficients:

$$\bar{V}_{ij} = 2\gamma M_{ij}, \qquad \bar{V}_{ij} = \left(\frac{\widetilde{C}_6}{2\gamma M_{ij}}\right)^{1/6}. \tag{13}$$

Because the target interaction strengths impose pairwise distances that need not admit an exact two-dimensional Euclidean embedding [10], we approximate them by numerically optimizing atom positions using sequential least-squares programming (SLSQP), subject to the device's minimum separation and maximum register radius. Embedding fidelity is quantified by

$$\epsilon_V = \sqrt{\frac{\sum_{i<j}(V_{ij} - \bar{V}_{ij})^2}{\sum_{i<j}(\bar{V}_{ij})^2}}. \tag{14}$$

3.2. Global Detuning and Pulse Design

The target diagonal coefficients are generally unequal, whereas the AnalogDevice supports only global detuning. Following the mean-detuning approximation used by Pasqal's QUBO solver [11], we choose

$$\delta_f = -\frac{\gamma}{N}\sum_i M_{ii}. \tag{15}$$

We adopt an adiabatic-inspired pulse schedule, following the baseline recommended in the challenge statement and Pasqal's Pulser documentation [1, 9]. The Rabi amplitude rises

smoothly from zero and returns to zero, enabling transitions between configurations during evolution, while the detuning sweeps from an initially negative value toward the target Hamiltonian.

$$\begin{aligned} \Omega(t) &= \Omega_p \sin^2\left(\frac{\pi t}{\tau}\right), \\ \delta(t) &= \delta_0 + (\delta_f - \delta_0)\frac{t}{\tau}, \\ 0 &\le t \le \tau. \end{aligned} \tag{16}$$

Rather than optimizing the pulse parameters, we determine them heuristically from the QUBO coefficients:

$$\begin{aligned} \Omega_p &= \mathrm{median}_{i<j}(2\gamma M_{ij}), \\ \delta_0 &= -\gamma \max_i |M_{ii}|, \\ \tau &= 4000 \text{ ns}. \end{aligned} \tag{17}$$

The median interaction provides a representative Rabi scale [8], while the negative initial detuning favors the unexcited register. The duration is a fixed baseline within the device's 6000 ns limit. The common energy scale γ is automatically restricted by the device's maximum Rabi frequency, detuning range, and minimum atom separation, with a 10% safety margin.

These choices provide a reproducible, hardware-compatible baseline without requiring classical pulse optimization. The resulting sequences are simulated using Pasqal's emu-sv state-vector emulator.

4. Results and Discussion

4.1. Experimental Results

We evaluated the Tier S formulation across 20 certified instances using 1,000 measurements per instance, $\lambda = 0.1$, and fixed 4 μs pulses without parameter optimization. Classical exhaustive enumeration established proven optimal clustering and CVRP references.

Metric	Emu-SV	Uniform Random
Feasibility rate	96.47%	58.66%
Clustering reference probability	63.97%	13.10%
CVRP reference probability	27.93%	12.76%
Mean best model gap	0%	0%
Mean best CVRP gap	5.47%	0%

The quantum emulator recovered the optimal clustering on all 20 instances and the optimal CVRP solution on 17. Its mean relative interaction error was 17.03%, demonstrating that exact embedding is not necessary to obtain useful measurements.

Although random sampling achieved a 0% best CVRP gap, this is attributable to the small 16-bitstring solution space. Quantum emulation produced substantially higher probabilities of reaching both references, but these results do not demonstrate a quantum advantage.

Comparison with Random Sampling.
Although quantum emulation achieved substantially higher feasibility (96.47% versus 58.66%) and clustering-reference probability (63.97% versus 13.10%), uniform random sampling recovered the optimal CVRP solution on all 20 instances, compared with 17 for quantum emulation. This result is primarily attributable to the small Tier S search space of only $2^4 = 16$ possible assignments, which can be explored extensively with 1,000 random samples. Furthermore, minimizing the clustering objective does not necessarily minimize the final routed CVRP cost. Consequently, quantum emulation concentrates probability on geographically favorable clusterings, while random sampling explores alternative partitions that may yield shorter routes. Although the quantum method demonstrates stronger per-shot performance, these results do not establish a computational advantage over classical random sampling.

4.2. Variable Efficiency and Scalability

The conventional formulation introduces binary assignment variables x_{ik}, assignment penalties, and binary-encoded capacity slack s_k:

$$\begin{aligned} F_{\text{std}} = C(x) &+ A\sum_i \left(\sum_k x_{ik} - 1\right)^2 \\ &+ B\sum_k \left(\sum_i d_i x_{ik} + s_k - Q\right)^2. \end{aligned} \tag{18}$$

The resulting variable requirements are compared below.

Encoding	Variable count	S	M	L	Fits in 80 atoms?
Assignment variables only	NK	8	30	60	Yes
+ Binary-encoded slack	$NK + K\lceil \log_2(Q+1)\rceil$	14	45	80	Yes
Our compact formulation	$N\lceil \log_2 K\rceil$	**4**	**20**	**30***	Yes*

The Tier L value is a projected assignment-variable count, not a fully implemented quadratic formulation. The conventional Tier L model reaches the 80-atom limit exactly.

Our approach reduces the Tier S variable requirement by 71.4% relative to the conventional slack-variable formulation. However, a smaller register does not guarantee that its interactions can be accurately embedded. This is apparent when we attempted to assess an M-Tier model by embedding a 20 qubit array, as we were unable to achieve a feasible layout using neutral-atom emulation. This is likely due to the original Unbalanced Penalization proposal our penalty function being built around quantum annealing, not analog neutral-atom systems.

4.3. What Went Wrong

Before selecting our final formulation, we investigated two alternatives: a conventional slack-variable QUBO, which introduced additional atoms and dense couplings, and a direct capacity-violation penalty, which was not quadratic without further transformation.

We also attempted to extend our final formulation to Tier M using 20 atoms. Although adaptive penalty selection improved the ideal QUBO, the physical embedding remained inaccurate. Even with emulation-only detuning map modulation (DMM), interaction error reached 63.62%. The best tested

embedding produced only 0.5% feasible measurements and a 44.85% best model gap.

These experiments illustrate the central limitation of our approach: variable reduction alone does not ensure physical embeddability. The target interaction matrix must also be compatible with the geometry and control restrictions of the neutral-atom device.

Ground-State Preservation. A central limitation of our implementation is the approximation introduced when mapping the target QUBO onto the physical Hamiltonian. Although the target QUBO retained exclusively clustering-optimal ground states on 19/20 instances, this condition held for only 12/20 instances after replacing the unequal diagonal coefficients with their mean global detuning. Following geometric embedding, it held for 14/20 instances.

This distinction strongly correlated with quantum sampling performance. The 14 instances preserving clustering-optimal embedded ground states achieved a median reference-reaching probability of 90.85%, compared with just 2.45% for the remaining six. Meanwhile, feasibility rates remained approximately 96% in both groups. Consequently, the overall 63.97% mean reference probability obscures a pronounced bimodal distribution. These findings suggest that preserving the target ground-state structure is more informative for predicting solution quality than embedding error alone.

4.4. AI Disclosure

Generative AI, primarily Claude, ChatGPT, and GrokBot, assisted with literature research, mathematical derivations, python implementation, debugging, experimental analysis, and report drafting. All reported numerical results were obtained from classical solvers or quantum emulators. The team reviewed the mathematical formulation and implementation and assumes responsibility for the accuracy of the submission.

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